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[Magby/CV](#)

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PROFILE

I am physics graduate from Valparaíso, Chile, with a specialization in scientific computational physics, particularly in simulations using high-order numerical methods and CUDA optimization. I have experience working with SQL, machine learning and deep learning with libraries as tensorflow and sklearn. I realized personal projects implementing CNN, RNN and PINN models for regression, classification, and clustering tasks. I also learned excel and power bi basic to intermediate level. I am continuously expanding my skill set by exploring new tools and concepts, including data warehousing..

Experience

- Undergraduate Thesis – "Simulation and Study of the Gravitational Collapse of a Massless Scalar Field with a Quantum Correction"
Universidad de Valparaíso • 07/2024 – 01/2025
 - Developed a numerical code to simulate the dynamics of a scalar field with quantum corrections and its effect on the space-time metric..
 - I learned advanced topics not covered in the university curriculum, such as General Relativity, classical and quantum field theory, the ADM formalism, and ghost fields. Studied and implemented high-order numerical methods, including 10th-order finite differences, 10th-order Runge-Kutta, and Kreiss-Oliger dissipation. Proposed custom boundary conditions that improved the stability of the simulation over time.
 - Wrote the code in C, optimized with CUDA. Data analysis and visualization were done in Python (matplotlib).
 - Received a research scholarship from the Centro de Física Teórica de Valparaíso (CeFiTeV) to support the thesis project.
- Teaching Assistant – Differential Equations • Facultad de ciencias, Universidad de Valparaíso,
Valparaíso • 4/2022 - 07/2022 & 04/2023 - 07/2023
 - Supported undergraduate Physics students by guiding them through the theory and solving exercises.

- Teaching Assistant – Thermodynamics• Facultad de ciencias, Universidad de Valparaíso, Valparaíso•08/2023 - 12/2023
 - Assisted students in understanding the laws of thermodynamics and solving related problems.
- Teaching Assistant – Intermediate Mechanics• Facultad de ciencias, Universidad de Valparaíso, Valparaíso•04/2024 - 07/2024
 - Assisted students solve complex mechanical systems using Lagrangian and Hamiltonian formalism.

Education

- Bachelor of Science in Physics, Universidad de Valparaíso, Valparaíso, Chile (2019-2024)
 - Specialized in scientific computing:
 1. Unix, Numerical Methods, Databases, Artificial Intelligence, Parallel Programming, GPU Optimization and Neural Networks.

Skills

- **Programming Languages:** Python (Intermediate) y C (Intermediate).
- **Data & Analysis:** SQL (Intermedio) y pandas (Intermedio), matplotlib y seaborn..
- **Machine learning and deep learning:** Tensorflow-Keras y scikit -learn (PINN, CNN y RNN, Random forest, SVM, Knn, etc...).
- **Numerical Libraries:** Numpy y scipy
- **Basic GUI app development:** TKinter (Python)
- **Scientific & High Performance Computing:** CUDA, MPI y OpenMP.
- **Microsoft:** Excel (Intermediate) y Power BI (basic)

Languages

- Español (nativo)
- Inglés (Basico) (A2)
 - Reading: Intermediate
 - Listening/Speaking: Basic